PHYSICS – BOARD QUESTION PAPER : FEBRUARY 2023

Time: 3 Hours

Max. Marks: 70

General Instructions:

The question paper is divided into four sections.

1. Section A: Q1 contains Ten multiple choice questions. Q2 contains Eight very short questions.
2. Section B: Q3 to Q14 contain Twelve questions. Attempt any Eight.
3. Section C: Q15 to Q26 contain Twelve questions. Attempt any Eight.
4. Section D: Q27 to Q31 contain Five questions. Attempt any Three.
5. Use of log table is allowed. Use of calculator is not allowed.
6. Figures to the right indicate full marks.
7. For MCQs, it is mandatory to write the correct answer along with its alphabet. Only first attempt is valid.

Physical Constants:

$h = 6.63 \times 10^{-34}$ Js

$c = 3 \times 10^{8}$ m/s

$\pi = 3.142$

$g = 9.8$ m/s²

$\varepsilon_0 = 8.85 \times 10^{-12}$ C²/Nm²

$\mu_0 = 4\pi \times 10^{-7}$ Wb/A–m

SECTION – A

Q.1 Select and write the correct answers: (10M)

i. Mean free path λ is:

(A) $2/\pi nd^2$ (B) $1/2\pi nd^2$ (C) $1/\pi nd^2$ (D) $1/2nd^2$

ii. First law of thermodynamics is consistent with conservation of:

(A) momentum (B) energy (C) mass (D) velocity

iii. Y = A + B is Boolean expression for:

(A) OR gate (B) AND gate (C) NOR gate (D) NAND gate

iv. Property of light unchanged from one medium to another:

(A) velocity (B) wavelength (C) amplitude (D) frequency

v. Circular coil 100 turns, area 1 m^2, B = 1 T. Magnetic flux =
(A) 1 Wb (B) 50 Wb (C) 100 Wb (D) 200 Wb

vi. For liquid that completely wets container:
(A) $\theta = 0$ (B) $0 < \theta < \pi/2$ (C) $\theta = \pi/2$ (D) $\pi/2 < \theta < \pi$

vii. LED emits when:
(A) junction reversed (B) depletion widens
(C) electrons & holes recombine (D) junction becomes hot

viii. Soft iron is used for transformer core due to:
(A) low coercivity & low retentivity
(B) low coercivity & high retentivity
(C) high coercivity & high retentivity
(D) high coercivity & low retentivity

ix. KE_{max} = 2 eV → stopping potential:
(A) 0.5 V (B) 1.0 V (C) 1.5 V (D) 2.0 V

x. Radius of 8th orbit is greater than 4th by factor:
(A) 2 (B) 4 (C) 8 (D) 16

Q.2 Answer the following: (8M)

i. Value of resistance of ideal voltmeter.
ii. Force on closed circuit in magnetic field.
iii. Average value of AC over one cycle.
iv. Electron accelerated through 100 V. Find de Broglie wavelength (nm).
v. If friction is zero on road, can vehicle move safely?
vi. Formula relating E and potential gradient.
vii. For $x = 5 \sin(\pi t/3)$, find velocity at t = 1 s.
viii. Write formula for Bohr magneton for nth orbit.

SECTION – B

Attempt any Eight questions. (16M)

Q.3. Define coefficient of viscosity. State formula and SI unit.

Q.4. Derive magnetic induction of toroid of N turns.

Q.5. State and prove conservation of angular momentum.

Q.6. Derive equivalent capacitance for series combination of C_1 and C_2.

Q.7. Why equivalent inductance of parallel coils is less than individual inductances?

Q.8. Explain how galvanometer is converted into ammeter.

Q.9. A 100Ω resistor connected to 220V, 50 Hz. Find (i) Irms (ii) Power consumed.

Q.10. Bar magnet 120 g, dimensions 40 mm × 100 mm × 80 mm. Period = π sec, moment = 3.4 Am^2. Find magnetic field.

Q.11. Distinguish free and forced vibrations (two points).

Q.12. Compare rate of heat loss at 827°C and 427°C (surroundings 27°C).

Q.13. Ideal monoatomic gas compressed adiabatically to twice temperature. Find pressure ratio.

Q.14. Activity 10^{10}/hr at t = 20 hr becomes 5×10^{9}/hr at t = 30 hr. Find decay constant.

SECTION – C

Attempt any Eight questions. (24M)

Q.15. Derive laws of reflection using Huygens' principle.

Q.16. State postulates of Bohr's atomic model.

Q.17. Define, state unit and dimensions of: (i) magnetization (ii) susceptibility.

Q.18. With circuit diagram, describe experiment of photoelectric effect.

Q.19. Explain determination of internal resistance using potentiometer.

Q.20. Explain working of n-p-n transistor in common base mode.

Q.21. Write differential equation of SHM. Derive acceleration and velocity expressions.

Q.22. Two tuning forks 320 Hz & 340 Hz. Velocity 326.4 m/s. Find difference in wavelength.

Q.23. Biprism: distance to eyepiece 1.2 m, 20th fringe at 0.4 cm. With convex lens 90 cm away, virtual images 0.9 cm apart. Find wavelength.

Q.24. Two long parallel wires distance 1.35 cm, force per unit length 4.76×10^{-2} N/m. Find current.

Q.25. AC voltage e = 140 sin(314.2 t) across 50Ω resistor. Find (i) frequency (ii) Irms.

Q.26. Dipole with charges ±1 μC separated by 2 cm in E = 10^5 N/C. Find (i) maximum torque (ii) work done in rotating 180°.

SECTION – D

Attempt any Three questions. (12M)

Q.27. Using kinetic theory derive expression for pressure of gas.

Q.28. (i) Derive energy stored in magnetic field.

(ii) A 5 m wire falls from 15 m height E–W direction. Find induced emf ($g = 10\ m/s^2$).

Q.29. (i) Derive work done in isothermal process.

(ii) 10^4 J work done on gas. Gas releases 125 kJ heat. Find change in internal energy.

Q.30. (i) Derive relation between surface energy and surface tension.

(ii) Work done in blowing bubble radius 1 cm ($\sigma = 2.5\times10^{-2}$ N/m).

Q.31. Derive expressions for linear velocity at lowest, mid-way and top-most positions for particle just completing vertical circular motion.